
I Am Speaking Into a Machine That Is Listening: Iterated Transcription as an Uncredited Co-Writer

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Abstract

Speech recognition never transcribes a performance faithfully. It mishears, drops inflection, and silently corrects grammar. We make those errors the agent. A spoken-word score is performed by a synthetic voice, transcribed, and the divergent transcript becomes the score for the next performance, iterated after Alvin Lucier’s *I Am Sitting in a Room* (1969). We measure the drift and ask whether speech reproduces the published finding that generative image loops collapse into a shared house style. Across eight scores in different registers (101 iterations, one commercial transcriber, \$3.02), seven of eight reach a fixed point close to their own score: final drift is 0.000 to 0.142 word error rate, and 93 to 100 percent of every final text remains the human’s words. The headline hypothesis fails. Pairwise vocabulary overlap between finals changes by +0.0019 against a per-pair standard deviation of 0.0094, only 13 of 28 pairs end closer than they started, and no word is shared by every final and absent from every seed. There is no shared attractor at this scale. What we find instead is sharper: the transcriber is nearly an identity function on its own output, but what it does write, it writes permanently and silently, at a constant 0.98 confidence that never dips at a substitution. The loop is not erosion but a few irreversible decisions taken early, with no signal that any decision was made. Transcription bias is an author nobody credits: small (0 to 7 percent of the final text), silent, and permanent. The piece is that ledger. It runs on a laptop with one API key in under two hours, with no participants.

1 Introduction

In 1969 Alvin Lucier sat in a room, recorded himself describing what he was doing, played the tape back into the room, and recorded that, again and again, until the room’s resonant frequencies ate his speech and only the room remained [Lucier, 1969]. The piece works because the apparatus is honest about its authorship: every pass, the room signs its name a little louder.

Modern speech recognition performs the same kind of rewriting and hides it. When a recognizer is uncertain it does not say so. It picks the likelier word and hands back a clean, fluent, confidently punctuated line. Every transcript is a collaboration billed to one party. Documented failures of this kind are not exotic: recognition error rates differ sharply by speaker demographics [Koenecke et al., 2020], and production transcription systems fabricate fluent text that was never spoken [Koenecke et al., 2024]. The fabrications read as language because they are drawn from the model’s language prior, which is exactly what makes them invisible.

This work turns that hidden co-writer into the entire piece. A spoken-word score is performed by a synthetic voice, transcribed, and the transcript, errors and all, becomes the score for the next performance:

$$\text{score} \rightarrow \text{performance} \rightarrow \text{mishearing} \rightarrow \text{score} \rightarrow \dots$$

Lucier’s room is replaced by a speech recognition model, and what it resonates at is its own language prior. The loop runs until the text stops moving, and then we ask two questions. Empirically: do texts fed through the same transcriber converge toward a common house style, the way self-consuming generative image loops collapse toward low-diversity attractors [Alemohammad et al., 2024, Shumailov et al., 2024]? Aesthetically: whose voice does the loop settle into, and what does the machine’s share of the final text look like when it is itemized word by word?

Our contributions are a runnable apparatus, a measurement, and a negative result reported as such. The apparatus is a self-contained loop over one commercial speech API, with per-iteration metrics, convergence detection, and an authorship accounting that splits every token of the final text into words the human wrote and words that entered through a mishearing and survived. The measurement is a sweep of eight scores across different registers, 101 iterations total. The negative result is that the image-domain collapse finding does not transfer at this scale: each score converges to its own private fixed point, not to a shared one. The positive finding, and the substance of the piece, is the character of the machine’s edits: rare, early, irreversible, and delivered at constant 0.98 confidence, so that certainty carries no information about correctness.

2 Related work

Process music and the tape loop. *I Am Sitting in a Room* [Lucier, 1969] is the lineage this work claims directly. Lucier’s insight was to let the recording apparatus compose by iteration, so that the final object is a portrait of the medium rather than the performer. We keep the procedure and swap the medium: the resonant filter is now a statistical model of English.

Self-consuming generative loops. Training or generating recursively on model output degrades diversity and pulls samples toward the model’s prior. Alemohammad et al. [2024] show image models entering *model autophagy disorder*, with samples collapsing in quality or diversity across generations, and Shumailov et al. [2024] demonstrate model collapse when models train on recursively generated data, with the tails of the original distribution disappearing first. These results motivate our empirical question, but the setups differ in an important way: those loops retrain or regenerate from the model’s distribution, while ours repeatedly applies a fixed recognize-and-resynthesize map to a text. Whether that map contracts different texts toward a common attractor is exactly what we test.

ASR bias and hallucination. Koenecke et al. [2020] document large racial disparities in commercial recognizers’ error rates, establishing that transcription error is structured rather than random. Koenecke et al. [2024] find that a widely deployed system invents whole phrases, including violent and fabricated content, in fluent register. Large weakly supervised recognizers [Radford et al., 2023] inherit strong language priors that improve average accuracy while making errors more fluent and therefore harder to see. Our loop is a device for making that structure visible: run the map until it stops moving, and its preferences are the residue.

3 The apparatus

The loop. One iteration performs the current score with a text-to-speech voice (Deepgram Aura 2, voice `aura-2-thalia-en`), transcribes the audio with a speech recognizer (Deepgram Nova 3, `smart_format on`), and adopts the transcript as the next score. Both halves of the loop are served by one API, so the whole piece runs on a laptop with a single key. The full sweep reported here took roughly 100 minutes and \$3.02. No human edits any intermediate text.

Where the performance lives. The load-bearing detail is breath grouping. The score is split at its punctuation, each phrase is synthesized separately, and real silence is inserted between phrases, so the phrasing exists in the waveform, where the transcript has no column for it. The other side of the same point is `smart_format`, the recognizer’s silent editor: it repunctuates, recapitalizes, and reformats numerals, never asked and never logged. Its output is treated as the transcript, exactly as a downstream user would treat it.

Scores. Eight original texts in Lucier’s lineage, written for this piece across deliberately different registers, so that a convergence claim would be a claim about the transcriber rather than about

Table 1: Per-score convergence. Drift is word error rate of the final text against the original score. Machine share is the fraction of final tokens that entered through a mishearing.

Score	Iterations	Outcome	Fixed point at	Drift vs. score	Machine share
credits	12	fixed point	7	0.039	3.3%
dialogue	3	fixed point	2	0.000	0.0%
form	25	open	–	0.065	4.1%
litany	9	fixed point	8	0.023	1.6%
lucier	25	fixed point	22	0.142	7.0%
plain	5	fixed point	2	0.000	0.0%
technical	9	fixed point	3	0.019	1.9%
verse	13	fixed point	10	0.104	3.9%

Table 2: Cross-run convergence over all 28 pairs of scores. The overlap change is one fifth of its own dispersion and should not be read as convergence.

Quantity	Seeds	Finals	Change
Mean pairwise vocabulary overlap	0.1420	0.1439	+0.0019
Mean pairwise cosine	0.529	0.517	−0.012
Pairs that ended closer		13 of 28	
Per-pair change, standard deviation		0.0094	
Words in every final, in no seed		0	

one text: a procedural self-description (*lucier*), argumentative prose (*credits*), performance instructions (*litany*), a two-speaker dialogue (*dialogue*), a form read aloud (*form*), lineated verse with deliberate homophones (*verse*), clinical exposition (*technical*), and unstructured hedging speech (*plain*).

Measurement. Each iteration records word and character error rate against the previous take and against the original score, vocabulary overlap and bag-of-words cosine against the score, token count, type-token ratio, punctuation density, and the recognizer’s own mean word confidence. Convergence is classified as *fixed point* (consecutive takes identical), *cycle*, or *open*. Authorship is counted over the final text’s tokens: the share that are words the human wrote, versus words that entered through a mishearing and were then re-performed and re-heard until they stuck. Across runs, we test for a shared attractor by comparing mean pairwise similarity between the eight seeds against mean pairwise similarity between the eight finals, reported with its dispersion.

4 Results

Eight scores, one transcriber, 101 iterations, 12 to 25 iterations per score with early stopping at a fixed point.

4.1 Each score finds its own fixed point

Seven of eight scores reach a fixed point, most within ten iterations (Table 1, Figure 1), and the fixed point is close to the score they started from: final drift ranges from 0.000 to 0.142 WER, and between 93 and 100 percent of every final text is still the words the human wrote. The transcriber is close to an identity function on its own output. Only *form*, the score built from numerals and list structure, was still moving at iteration 25; its final state is a truncation of the experiment, not a fixed point. Figure 2 shows the longest and most mobile run, *lucier*, whose drift accumulates over 22 iterations before locking.

4.2 No shared attractor

The headline hypothesis, that different scores pushed through the same transcriber would drift toward a common house style the way self-consuming image loops do [Alemohammad et al., 2024,

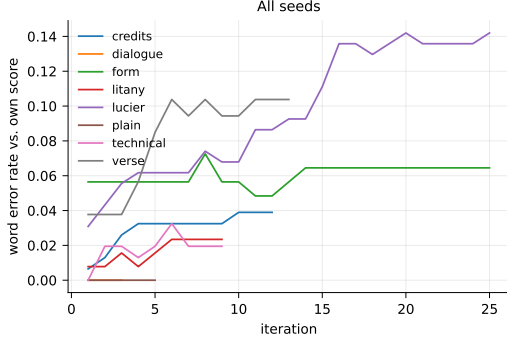


Figure 1: All eight scores, WER against their own original score, per iteration. Each score climbs briefly and locks at its own level. The curves do not approach one another.

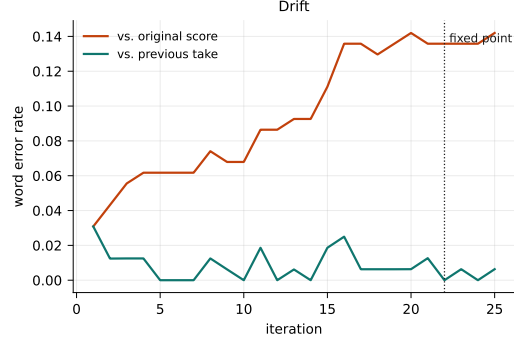


Figure 2: The *lucier* run. WER against the previous take falls to zero as the loop locks, while distance from the score accumulates and holds. The dotted line marks the fixed point at iteration 22.

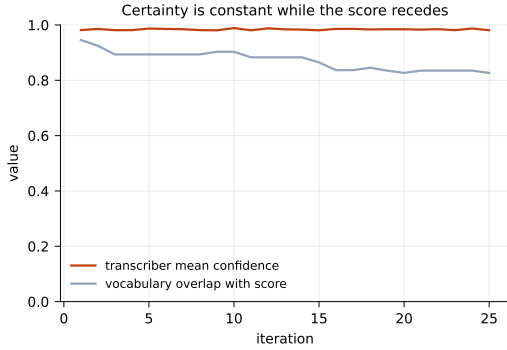


Figure 3: The *lucier* run. Mean word confidence is constant at roughly 0.98 while vocabulary overlap with the score falls to 0.83. Certainty carries no information about how far the text has traveled.

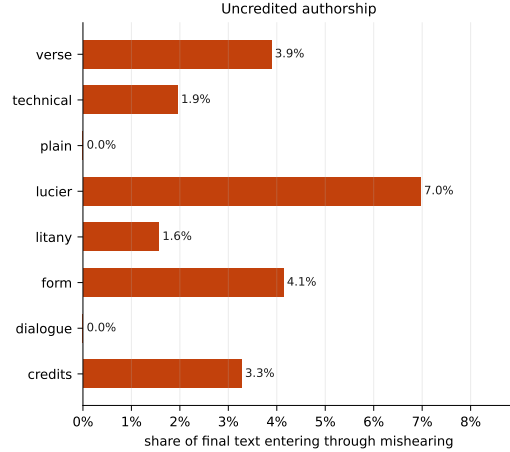


Figure 4: Uncredited authorship. For each score, the share of the final text’s tokens that entered through a mishearing and survived to the end of the run.

Shumailov et al., 2024], is not supported (Table 2). Mean pairwise vocabulary overlap between the eight final texts moves by +0.0019, one fifth of the per-pair standard deviation of 0.0094. Mean pairwise cosine moves the wrong way, from 0.529 to 0.517. Only 13 of 28 pairs end closer together than they started, which is worse than chance. No word appears in every final text while being absent from every seed, so there is no shared attractor vocabulary at all. Each score converges to its own private fixed point. We report this as a negative result: at this scale, with this transcriber, the image-domain collapse finding does not transfer to iterated transcription.

Two secondary predictions also failed. Confidence does not rise as the text drifts; it is flat at roughly 0.98 throughout (Figure 3), because the transcriber was maximally certain at iteration one and had nowhere to go. And the performance channel does not collapse: punctuation per 100 tokens rises on *lucier* from 10.6 to 12.7, type-token ratio rises slightly, and token counts barely move (154 to 153, 128 to 128, 105 to 103). *smart_format* reformats aggressively at every step, but it does not strip structure away over time.

4.3 What the machine writes, it writes permanently

The interesting behavior is not erosion but a small number of silent, irreversible decisions taken early. Surviving substitutions are dominated by homophone and near-homophone swaps (*here* for *hear*, *not* for *nought*, *planar* for *plainer*), followed by contraction (*i'm* for *am*) and numeral reformatting (*11* for *eleven*). Each is a coin the language prior flipped once and never reconsidered; mean confidence never dips at a substitution.

The clearest single instance is the last line of the *lucier* score, whose fixed point at iteration 22 reads against its origin as follows:

```
score: I am speaking into a machine that is listening.
final: I'm speaking it to a machine that is listening.

score: ...to hear the way the machine smooths what it cannot place.
final: ...to hear the way machines smooth, but it cannot place.

score: You will hear my sentences give way to its sentences.
final: You will hear my sentences give way to sentences.

score: The mishearings are the author now.
final: The mishearings are he offer now.
```

Possessives are lost, clauses are fused, and a negation-adjacent substitution (*smooths what it cannot place* to *smooth, but it cannot place*) changes the sentence's meaning while reading as fluent English. On this run, the words that entered through mishearing and survived to the fixed point are *i'm*, *offer*, *towards*, *nobody's*, *out*, *but*, *machines*, *smooth*; the score's own words never heard again are *author*, *into*, *its*, *nobody*, *on*, *smooths*, *toward*, *weight*. The word *author* is, literally, one of the eight words this transcriber could not keep.

5 Whose voice

The aesthetic claim of the piece is that transcription bias is an author nobody credits, and the measurement puts three adjectives on that co-writer. It is *small*: 0 to 7 percent of the final text (Figure 4). It is *silent*: 0.98 confidence at every substitution, so the interface offers no seam between the human's words and the machine's. And it is *permanent*: every surviving substitution persists to the fixed point, because the recognizer reproduces its own output faithfully even when that output was wrong.

This reframes the negative result. The loop does not settle into the machine's voice wholesale, the way *Lucier's* room eventually spoke alone. It settles into the human's voice with the machine's edits fused invisibly inside it, which is arguably the more unsettling attractor. *Lucier's* room announced itself; this room forges the performer's signature. The final texts are collaborations in which the junior author's contributions can only be recovered by keeping the ledger this apparatus keeps, and no deployed transcription interface keeps that ledger.

As a creative system, the piece is fully reproducible: the loop, the metrics, the figures, and the eight scores are released, the sweep costs about three dollars, and a `-seed-audio` flag accepts a human reading in place of the synthetic performer, so the work can be restaged with any voice, any language the recognizer claims to support, and any future transcriber, each of which will sign its own name differently.

6 Limitations

One transcriber, one synthetic voice, one language: the negative convergence result is about Nova 3 with Aura 2, not about speech recognition generally. Eight scores yield 28 pairs, which is underpowered for a small cross-run effect, and that is precisely why the +0.0019 change must not be read as a positive. The performances are synthetic throughout; a human reader would introduce disfluency and prosody that the TTS voice does not produce. Both halves of the loop are nondeterministic: an earlier partial sweep reached the `credits` fixed point at iteration 4 where this sweep reached it at 7, so convergence iteration is not stable across runs, though convergence itself occurred in both. Finally, form did not converge within 25 iterations, so its reported final

state is a truncation rather than a fixed point, and the most obvious unrun experiment is the ablation of `smart_format`, which would isolate one feature’s authorship by diffing the two attractors it produces.

7 Conclusion

We iterated the transcription map until it stopped moving and read what it stopped at. The speech domain, at this scale, does not reproduce the image domain’s collapse into house style: each text keeps its own attractor. What the loop exposes instead is the texture of machine authorship in transcription: a handful of confident, early, permanent rewrites that are indistinguishable from the human’s text without an audit trail. The mishearings are the author now, or at least a co-author, and the piece exists so that, for once, they are credited.

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References

- Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoochi, and Richard G. Baraniuk. Self-consuming generative models go MAD. In *International Conference on Learning Representations (ICLR)*, 2024.
- Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R. Rickford, Dan Jurafsky, and Sharad Goel. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689, 2020.
- Allison Koenecke, Anna Seo Gyeong Choi, Katelyn X. Mei, Hilke Schellmann, and Mona Sloane. Careless whisper: Speech-to-text hallucination harms. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024.
- Alvin Lucier. *I Am Sitting in a Room*. Sound composition for voice and tape, 1969.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*, 2023.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.